Recovering Coordinate Signature Characteristic Functions from Unitary Path Characteristic Functions

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Abstract

Let X be a random continuous path of bounded variation in $\mathbb{R}^d$, and let $S(X)$ be its signature. The unitary path characteristic function is the family

$$\Phi_X(M) := \mathbb{E}[\mathrm{Dev}_M(X)], \qquad M \in \mathrm{Hom}(\mathbb{R}^d, \mathfrak{u}_N),\ N \geq 1,$$

where $\mathrm{Dev}_M(X)$ is the endpoint of the unitary development driven by X. For a word I and a real number λ, we study the recovery of

$$\varphi_I(\lambda) := \mathbb{E}\exp\big(i\lambda\,\pi_I(S(X))\big)$$

from Φ_X. The recovery is required not to proceed by first reconstructing $\pi_I(S(X))$ pathwise and then applying the exponential. Instead, $\exp(i\lambda\pi_I)$ is treated as an infinite shuffle-linear functional on signatures. Under an explicit exponential-integrability condition, we give an exact inversion formula. The m-th moment of $\pi_I(S(X))$ is recovered as a single Taylor coefficient of Φ_X evaluated on a finite-dimensional tensor power of a unitary word-picker. Hence

$$\varphi_I(\lambda) = \sum_{m=0}^{\infty} \frac{(i\lambda)^m}{m!}\,\frac{1}{(|I|m)!}\,\frac{\mathrm{d}^{|I|m}}{\mathrm{d}t^{|I|m}}\Big|_{t=0} \big[\Phi_X(tM_I^{[m]})\big]_{\mathbf{0},\mathbf{k}},$$

with the terms defined explicitly below. Each coefficient uses only a finite unitary group, although the characteristic function is generally an infinite series.

1 Problem and admissible information

Throughout, paths are assumed continuous and of finite 1-variation. This continuity convention is important: for a discontinuous bounded-variation path, the equation $\mathrm{d}Y = Y\,M(\,\mathrm{d}X)$ need not preserve unitarity across jumps under the ordinary Lebesgue–Stieltjes interpretation. The same algebraic formulas remain valid under any jump convention for which the usual signature expansion of the development is valid and skew-Hermitian coefficients give unitary increments.

For a deterministic path $x : [0,T] \to \mathbb{R}^d$ and a word $w = (j_1, \ldots, j_n)$, write

$$\pi_w(S(x)) := \int_{0<t_1<\cdots<t_n<T} \mathrm{d}x_{t_1}^{j_1} \cdots \mathrm{d}x_{t_n}^{j_n}, \qquad \pi_\varnothing(S(x)) := 1.$$

The signature is the formal series of these iterated integrals. We use the standard geometric, or Chen, convention; see [Che57; LCL07].

Let U_N be the unitary group on $\mathbb{C}^N$ and

$$\mathfrak{u}_N = \{A \in \mathbb{C}^{N\times N} : A^* = -A\}.$$

Given $M \in \mathrm{Hom}(\mathbb{R}^d, \mathfrak{u}_N)$, the development of x under M is the solution of

$$\mathrm{d}Y_t = Y_t\, M(\,\mathrm{d}x_t), \qquad Y_0 = I_N,$$

and $\mathrm{Dev}_M(x) := Y_T$. For a random path X, its unitary path characteristic function is

$$\Phi_X(M) := \mathbb{E}[\mathrm{Dev}_M(X)].$$

This is the object that is assumed to be known, for all finite N and all skew-Hermitian coefficient maps M. This terminology is consistent with the rough-path characteristic functions of Chevyrev and Lyons [CL16].

The stochastic target is not the deterministic value $\exp(i\lambda\pi_I(S(x)))$ for an observed path x, but rather

$$\varphi_I(\lambda) := \mathbb{E}\exp\big(i\lambda\,\pi_I(S(X))\big).$$

This distinction is essential. In the deterministic problem one may recover $\pi_I(S(x))$ and then exponentiate it. In the stochastic problem this shortcut has no analogue: recovering $\mathbb{E}\pi_I(S(X))$ gives only the first moment and does not determine $\mathbb{E}e^{i\lambda\pi_I(S(X))}$.

Remark 1 (A minimal warning example). Let $d = 1$ and $X_t = tZ$ on $[0,1]$, where $Z = 1$ and $Z = -1$ with probability $1/2$ each. Then $\pi_{(1)}(S(X)) = Z$, so

$$\mathbb{E}\pi_{(1)}(S(X)) = 0, \qquad \mathbb{E}e^{i\lambda\pi_{(1)}(S(X))} = \cos\lambda.$$

Exponentiating the mean would give 1, not $\cos\lambda$.

We therefore impose the following information constraint.

Definition 1 (No pathwise exponential-inversion shortcut). An admissible recovery procedure may use only the map $M \mapsto \Phi_X(M)$, algebraic operations on finite-dimensional matrices, Taylor coefficients of scalar entries of $t \mapsto \Phi_X(tM)$ at $t = 0$, and convergent scalar series. It may not assume access to individual sample-path developments $\mathrm{Dev}_M(X(\omega))$, may not reconstruct the random variable $\pi_I(S(X))$ pathwise, and may not compute $\varphi_I(\lambda)$ by replacing it with $\exp(i\lambda\,\mathbb{E}\pi_I(S(X)))$.

The construction below satisfies this constraint. It recovers the moments $\mathbb{E}[\pi_I(S(X))^m]$ as linear functionals of the law of X, and then sums the ordinary characteristic-function series under a regularity assumption that justifies the exchange of the infinite series and expectation.

2 Two algebraic preliminaries

2.1 Development as evaluation of the signature

If $A_1, \dots, A_d$ are matrices and $M(e_a) = A_a$, then the right-multiplicative linear equation has the signature expansion

$$\mathrm{Dev}_M(x) = I_N + \sum_{n=1}^{\infty} \sum_{j_1,\dots,j_n=1}^{d} \pi_{(j_1,\dots,j_n)}(S(x))A_{j_1}\cdots A_{j_n}. \tag{1}$$

The product order in (1) follows from the convention $\mathrm{d}Y = Y\,M(\,\mathrm{d}x)$. The series is absolutely convergent for bounded-variation paths. Indeed, with

$$V(x) := \sum_{a=1}^{d} |x^a|_{1\text{-var};[0,T]}, \qquad C_M := \max_{1\le a\le d} \|M(e_a)\|,$$

one has the estimate

$$\|\operatorname{Dev}_{zM}(x)\| \le \exp(|z|C_M V(x)), \qquad z \in \mathbb{C}. \tag{2}$$

Here zM is used only as an analytic proof device; the path characteristic function is queried only for real $z = t$, for which tM remains skew-Hermitian.
Hao: We should specify which matrix norm is used before stating the inequality Eqn. (2). Otherwise, this equality might not be true.

2.2 Shuffle linearity of exponentials of coordinates

Let ℓ_w denote the coordinate functional $\ell_w(S(x)) = \pi_w(S(x))$. The coordinate functionals satisfy Chen's shuffle identity

$$\ell_u(S(x))\ell_v(S(x)) = (\ell_u \sqcup\!\sqcup \ell_v)(S(x)), \tag{3}$$

where $\sqcup\!\sqcup$ denotes the shuffle product. Thus

$$\pi_I(S(x))^m = (\ell_I^{\sqcup\!\sqcup m})(S(x)).$$

[Ni: I think that this shuffle product property is not needed for the proof.] Consequently, formally,

$$\mathcal{E}_{I,\lambda} := \sum_{m=0}^{\infty} \frac{(i\lambda)^m}{m!}\, \ell_I^{\sqcup\!\sqcup m} \tag{4}$$

is a linear functional on a suitable completion of the tensor algebra, and

$$\mathcal{E}_{I,\lambda}(S(x)) = \exp\big(i\lambda\pi_I(S(x))\big).$$

The analytic issue is not algebraic but integrability: to pass from the formal identity to an identity after expectation, we need absolute convergence in L^1.

For a word I of length k, the elementary bound

$$|\pi_I(S(x))| \le \frac{V(x)^k}{k!} \tag{5}$$

will be used repeatedly.

3 Regularity assumption

Fix a nonempty word I of length k and a real number λ. We use the following condition.

Assumption 1 (Exponential variation moment). There exists $\alpha > |\lambda|/k!$ such that

$$\mathbb{E}\exp(\alpha V(X)^k) < \infty. \tag{$A_{I,\lambda}$}$$

If $V(X)$ is almost surely bounded, then Assumption 1 holds for every I and every λ. If one wants simultaneous recovery for all real λ for a fixed word length k, it is enough to assume $\mathbb{E}e^{\alpha V(X)^k} < \infty$ for every $\alpha > 0$.

Assumption 1 is stronger than mere existence of the characteristic function, since $e^{i\lambda\pi_I}$ is bounded and hence always integrable. Its role is to justify the shuffle-Taylor expansion

$$\mathbb{E}e^{i\lambda\pi_I(S(X))} = \sum_{m=0}^{\infty} \frac{(i\lambda)^m}{m!}\,\mathbb{E}[\pi_I(S(X))^m].$$

Without such an analytic moment condition, the characteristic function may still exist, but this particular exact inversion via the infinite shuffle-linear functional is not justified.

Lemma 1 (Analytic exchange near the origin)**.** *Suppose Assumption 1. Let $M \in \mathrm{Hom}(\mathbb{R}^d, \mathfrak{u}_N)$ be any fixed finite-dimensional coefficient map. Then the function*

$$z \longmapsto \mathbb{E}[\mathrm{Dev}_{zM}(X)]$$

is analytic in a neighbourhood of 0, and its Taylor coefficients are obtained by taking expectations of the corresponding pathwise Taylor coefficients.

Proof. By (2),

$$\|\,\mathrm{Dev}_{zM}(X)\| \le \exp(|z|C_M V(X)).$$

Since Assumption 1 gives an exponential moment of $V(X)^k$ with positive coefficient, it implies an exponential moment of $V(X)$ with sufficiently small positive coefficient. More explicitly, if $k = 1$, choose $|z|C_M < \alpha$; if $k \ge 2$, Young's inequality gives $|z|C_M V \le (\alpha/2)V^k + C(z, M, \alpha)$ for fixed small z. Hence $\exp(|z|C_M V(X))$ is integrable on a small disk around 0. The pathwise map $z \mapsto \mathrm{Dev}_{zM}(X)$ is entire by the absolutely convergent expansion (1). Dominated convergence, equivalently Cauchy's formula under the expectation, gives analyticity of the expectation and identifies the Taylor coefficients with the expectations of the pathwise coefficients. □

4 A unitary word-picker

Let $I = (i_1, \dots, i_k)$ with $k \ge 1$. Work on $\mathbb{C}^{k+1}$ with basis $e_0, e_1, \dots, e_k$. Let E_{pq} denote the matrix unit satisfying $E_{pq}e_q = e_p$. Define, for $r = 1, \dots, k$,

$$B_r := E_{r-1,r} - E_{r,r-1}.$$

Then $B_r^* = -B_r$, so $B_r \in \mathfrak{u}_{k+1}$. Set

$$C_a := \sum_{\substack{1 \le r \le k \\ i_r = a}} B_r, \qquad M_I(e_a) := C_a, \qquad a = 1, \dots, d. \tag{6}$$

Thus $M_I \in \mathrm{Hom}(\mathbb{R}^d, \mathfrak{u}_{k+1})$.

Lemma 2 (Lowest-order word extraction)**.** *For every deterministic path x of bounded variation,*

$$\big[\mathrm{Dev}_{tM_I}(x)\big]_{0,k} = t^k \pi_I(S(x)) + O_x(t^{k+1}) \qquad (t \to 0). \tag{7}$$

Equivalently,

$$\pi_I(S(x)) = \frac{1}{k!}\,\frac{\mathrm{d}^k}{\mathrm{d}t^k}\bigg|_{t=0} \big[\mathrm{Dev}_{tM_I}(x)\big]_{0,k}.$$

Proof. By (1),

$$\mathrm{Dev}_{tM_I}(x) = I_{k+1} + \sum_{n=1}^{\infty} t^n \sum_{r_1,\dots,r_n=1}^{k} \pi_{(i_{r_1},\dots,i_{r_n})}(S(x)) B_{r_1} \cdots B_{r_n}.$$

The matrix B_r connects only the adjacent vertices $r-1$ and r in the line $0-1-\cdots-k$. Therefore the entry $(B_{r_1}\cdots B_{r_n})_{0,k}$ can be nonzero only if there exists a nearest-neighbour walk from 0 to k in n steps. No such walk exists for $n < k$.

At $n = k$, the only nearest-neighbour walk from 0 to k is the monotone walk $0 \to 1 \to \cdots \to k$. Hence the only contributing edge sequence is $(r_1, \dots, r_k) = (1, \dots, k)$, and its coefficient is

$$(B_1 \cdots B_k)_{0,k} = 1.$$

The corresponding signature coordinate is $\pi_{(i_1,\dots,i_k)}(S(x))$. This proves (7). □

[Ni: We should cite my previous paper on restricted PCF.]

Remark 2 (Repeated letters). If the same alphabet letter appears in several positions of I, the matrices still distinguish the positions. For example, if $I = (a, a)$, then $C_a = B_1 + B_2$, but the product B_1B_2 contributes to the $(0, 2)$ entry while B_2B_1 does not. Thus repeated letters do not create ambiguity.

Lemma 2 is a deterministic extraction statement, but the stochastic algorithm below will not use it to reconstruct $\pi_I(S(X))$ pathwise. It will use its leading-term consequence only after taking tensor powers and expectations.

5 Recovering the moments without pathwise recovery

For $m \geq 1$, define the Hilbert space

$$H_m := (\mathbb{C}^{k+1})^{\otimes m}.$$

We index its standard basis by multi-indices $\mathbf{p} = (p_1, \dots, p_m) \in \{0, \dots, k\}^m$ and write

$$e_{\mathbf{p}} := e_{p_1} \otimes \cdots \otimes e_{p_m}.$$

Let

$$\mathbf{0} := (0, \dots, 0), \qquad \mathbf{k} := (k, \dots, k).$$

Define the tensor-lifted coefficient map

$$M_I^{[m]}(e_a) := \sum_{q=1}^{m} \mathrm{Id}^{\otimes(q-1)} \otimes M_I(e_a) \otimes \mathrm{Id}^{\otimes(m-q)}, \qquad a = 1, \dots, d. \tag{8}$$

Since each $M_I(e_a)$ is skew-Hermitian, each $M_I^{[m]}(e_a)$ is skew-Hermitian on H_m. Hence

$$M_I^{[m]} \in \mathrm{Hom}(\mathbb{R}^d, \mathfrak{u}_{(k+1)^m}).$$

Lemma 3 (Tensor powers of developments)**.** *For every deterministic bounded-variation path x,*

$$\mathrm{Dev}_{tM_I^{[m]}}(x) = \mathrm{Dev}_{tM_I}(x)^{\otimes m}. \tag{9}$$

Consequently,

$$\left[\mathrm{Dev}_{tM_I^{[m]}}(x)\right]_{\mathbf{0},\mathbf{k}} = t^{km}\pi_I(S(x))^m + O_x(t^{km+1}). \tag{10}$$

Proof. Let Y_t solve $\mathrm{d}Y_t = Y_t\, tM_I(\mathrm{d}x_t)$ with $Y_0 = I_{k+1}$. By the ordinary product rule for bounded-variation paths, $Z_t := Y_t^{\otimes m}$ satisfies

$$\mathrm{d}Z_t = Z_t\, t\sum_{q=1}^{m} \mathrm{Id}^{\otimes(q-1)} \otimes M_I(\mathrm{d}x_t) \otimes \mathrm{Id}^{\otimes(m-q)}, \qquad Z_0 = I_{H_m}. \tag{11}$$

This is precisely the development equation with coefficient map $tM_I^{[m]}$, proving (9). [Ni: for clarity, one should add tensor product property to prove Eqn. 11.] Taking the matrix coefficient from $e_{\mathbf{k}} = e_k^{\otimes m}$ to $e_{\mathbf{0}} = e_0^{\otimes m}$ gives

$$\left[\mathrm{Dev}_{tM_I^{[m]}}(x)\right]_{\mathbf{0},\mathbf{k}} = \left([\mathrm{Dev}_{tM_I}(x)]_{0,k}\right)^m.$$

By Lemma 2, the scalar entry $[\mathrm{Dev}_{tM_I}(x)]_{0,k}$ has lowest nonzero order k and leading coefficient $\pi_I(S(x))$. Raising it to the m-th power gives (10). □

Proposition 1 (Moment recovery from the path characteristic function)**.** *Suppose Assumption 1. For every $m \geq 1$,*

$$\mu_m(I) := \mathbb{E}[\pi_I(S(X))^m] = \frac{1}{(km)!}\,\frac{\mathrm{d}^{km}}{\mathrm{d}t^{km}}\bigg|_{t=0} \left[\Phi_X(tM_I^{[m]})\right]_{\mathbf{0},\mathbf{k}}. \tag{12}$$

Set $\mu_0(I) := 1$.

Proof. For fixed m, the coefficient map $M_I^{[m]}$ is finite-dimensional. By Lemma 1, the Taylor coefficients at zero of $t \mapsto \Phi_X(tM_I^{[m]})$ are the expectations of the corresponding pathwise Taylor coefficients. Lemma 3 identifies the coefficient of t^{km} in the $(\mathbf{0},\mathbf{k})$ entry as $\pi_I(S(X))^m$. Multiplying by $(km)!$ gives (12). □

This proposition is the central replacement for the deterministic shortcut. It does not recover a sample-path value of $\pi_I(S(X))$. It recovers each scalar moment $\mu_m(I)$ directly from the path characteristic function, using a finite-dimensional unitary query of dimension $(k+1)^m$.

6 Main inversion formula

Theorem 1 (Coordinate characteristic function from unitary path characteristic function)**.** *Let X be a random continuous bounded-variation path in $\mathbb{R}^d$. Let $I = (i_1, \ldots, i_k)$ be a nonempty word and let $\lambda \in \mathbb{R}$. Assume Assumption 1. Then $\varphi_I(\lambda) = \mathbb{E}e^{i\lambda\pi_I(S(X))}$ is recovered from the unitary path characteristic function Φ_X by the exact formula*

$$\boxed{\varphi_I(\lambda) = \sum_{m=0}^{\infty} \frac{(i\lambda)^m}{m!}\,\mu_m(I),} \tag{13}$$

where $\mu_0(I) = 1$ *and, for* $m \geq 1$,

$$\boxed{\mu_m(I) = \frac{1}{(km)!} \left.\frac{\mathrm{d}^{km}}{\mathrm{d}t^{km}}\right|_{t=0} \left[\Phi_X(tM_I^{[m]})\right]_{\mathbf{0},\mathbf{k}}.} \tag{14}$$

The series in (13) *is absolutely convergent.*

Proof. Let

$$Z_I := \pi_I(S(X)).$$

By (5),

$$|Z_I| \leq \frac{V(X)^k}{k!}.$$

Since Assumption 1 gives an exponential moment with parameter strictly larger than $|\lambda|/k!$, we have

$$\mathbb{E}e^{|\lambda||Z_I|} \leq \mathbb{E}\exp\left(\frac{|\lambda|}{k!}V(X)^k\right) < \infty.$$

Therefore the scalar exponential series is absolutely integrable:

$$\mathbb{E}\sum_{m=0}^{\infty}\frac{|\lambda|^m}{m!}|Z_I|^m = \mathbb{E}e^{|\lambda||Z_I|} < \infty.$$

Tonelli's theorem and dominated convergence imply

$$\mathbb{E}e^{i\lambda Z_I} = \mathbb{E}\sum_{m=0}^{\infty}\frac{(i\lambda)^m}{m!}Z_I^m = \sum_{m=0}^{\infty}\frac{(i\lambda)^m}{m!}\mathbb{E}[Z_I^m].$$

Proposition 1 expresses each moment $\mathbb{E}[Z_I^m]$ by the finite-dimensional derivative (14). Substitution gives (13). Absolute convergence follows from the displayed bound. □

For the empty word, $\pi_\varnothing(S(X)) = 1$, so no path information is needed:

$$\varphi_\varnothing(\lambda) = e^{i\lambda}.$$

For a word of length one, say $I = (a)$, there is an even simpler finite-dimensional formula that requires no moment assumption: choose the scalar skew-Hermitian map $M_\lambda : \mathbb{R}^d \to \mathfrak{u}_1 = i\mathbb{R}$ given by

$$M_\lambda(e_a) = i\lambda, \qquad M_\lambda(e_b) = 0 \quad (b \neq a).$$

Then

$$\Phi_X(M_\lambda) = \mathbb{E}e^{i\lambda\pi_{(a)}(S(X))}.$$

The theorem is most useful for words of length at least two, where the target is no longer itself a one-dimensional unitary development.

7 Equivalent shuffle-functional interpretation

The moment formula above is equivalent to applying the infinite shuffle-linear functional (4) to the expected signature. Indeed, by the shuffle identity (3),

$$\ell_I^{\sqcup\!\sqcup m}(S(X)) = \pi_I(S(X))^m.$$

Under Assumption 1, the series is absolutely integrable, so

$$\varphi_I(\lambda) = \mathbb{E}[\mathcal{E}_{I,\lambda}(S(X))] = \mathcal{E}_{I,\lambda}(\mathbb{E}S(X)). \tag{15}$$

Thus the nonlinear exponential in the scalar coordinate becomes a linear functional on the signature once one passes to the completed shuffle algebra. The expectation can then be exchanged with this linear functional precisely because the absolute-convergence condition holds.

One could evaluate (15) by recovering, one by one, all expected signature coordinates appearing in the shuffle powers $\ell_I^{\sqcup\!\sqcup m}$. The tensor-power construction in Proposition 1 is a more compact implementation: the m-th tensor-lifted unitary development extracts the whole shuffle moment $\mathbb{E}[\pi_I(S(X))^m]$ in one derivative, without enumerating all shuffles.

8 Algorithm

Given the path characteristic function Φ_X, a nonempty word $I = (i_1, \dots, i_k)$, and $\lambda \in \mathbb{R}$, proceed as follows.

(1) If $k = 1$, use the one-dimensional formula above. If $k \geq 2$, continue.

(2) Build the skew-Hermitian line-graph matrices

$$B_r = E_{r-1,r} - E_{r,r-1}, \qquad r = 1, \dots, k,$$

and the word-picker $M_I : \mathbb{R}^d \to \mathfrak{u}_{k+1}$ by

$$M_I(e_a) = \sum_{\substack{1 \leq r \leq k \\ i_r = a}} B_r.$$

(3) For each $m \geq 1$, build the finite-dimensional tensor lift $M_I^{[m]} : \mathbb{R}^d \to \mathfrak{u}_{(k+1)^m}$ by

$$M_I^{[m]}(e_a) = \sum_{q=1}^{m} \mathrm{Id}^{\otimes(q-1)} \otimes M_I(e_a) \otimes \mathrm{Id}^{\otimes(m-q)}.$$

(4) Query the scalar function

$$t \longmapsto [\Phi_X(tM_I^{[m]})]_{\mathbf{0},\mathbf{k}}$$

near $t = 0$, and compute its exact Taylor coefficient of degree km:

$$\mu_m(I) = \frac{1}{(km)!} \left. \frac{\mathrm{d}^{km}}{\mathrm{d}t^{km}} \right|_{t=0} [\Phi_X(tM_I^{[m]})]_{\mathbf{0},\mathbf{k}}.$$

(5) Return the absolutely convergent value

$$\varphi_I(\lambda) = \sum_{m=0}^{\infty} \frac{(i\lambda)^m}{m!} \mu_m(I), \qquad \mu_0(I) = 1.$$

The procedure is exact in the mathematical oracle sense. It is finite-dimensional for each moment, with dimension $(k+1)^m$ for the m-th moment. In general the final characteristic function is not a finite sum: exponentials of nontrivial coordinates have infinitely many nonzero shuffle powers. If one truncates the series for numerical computation, the remainder is controlled by

$$\left| \varphi_I(\lambda) - \sum_{m=0}^{R} \frac{(i\lambda)^m}{m!} \mu_m(I) \right| \le \mathbb{E} \sum_{m=R+1}^{\infty} \frac{(|\lambda||Z_I|)^m}{m!}, \qquad Z_I = \pi_I(S(X)).$$

For example, if $V(X) \le L$ almost surely, this is bounded by the corresponding scalar exponential tail with $|Z_I| \le L^k/k!$.

9 Checks and limitations

Remark 3 (The construction is linear in the law). Every quantity $\Phi_X(tM_I^{[m]})$ is an expectation of a unitary development, so it depends linearly on the law of X. The derivative in (14) extracts a linear functional of the expected signature. The only nonlinear scalar operation is the final summation of the characteristic function series; this summation is justified by Assumption 1 and is not a pathwise inversion of the exponential.

Remark 4 (Why the exponential-integrability assumption appears). The bounded random variable $e^{i\lambda Z_I}$ is always integrable. However, the identity

$$\mathbb{E}e^{i\lambda Z_I} = \sum_{m \ge 0} \frac{(i\lambda)^m}{m!} \mathbb{E}[Z_I^m]$$

requires analyticity at the origin, not merely existence of the characteristic function. Moment-indeterminate or heavy-tailed examples show that formal moment expansions cannot be used without a regularity condition. Assumption Assumption 1 is a simple sufficient condition adapted to bounded-variation signature estimates.

Remark 5 (Multiplication convention). The formulas use the right-multiplicative convention $\mathrm{d}Y = Y\, M(\,\mathrm{d}X)$. With the left-multiplicative convention $\mathrm{d}Y = M(\,\mathrm{d}X)Y$, the same edge construction would pick the reversed word unless the matrix ordering were adjusted.

Remark 6 (Relation to known uniqueness results). Expected signatures and characteristic functions of rough paths are known to carry law-determining information under suitable hypotheses; see, for example, [HL10; CL16]. The result here is more local and constructive: for a prescribed word I, it gives an explicit coefficient-level procedure for recovering the one-dimensional characteristic function of the coordinate random variable $\pi_I(S(X))$ from the unitary path characteristic function.

10 Conclusion

In the deterministic problem, one can recover $\pi_I(S(x))$ and then apply the ordinary scalar exponential. That shortcut is not the right stochastic principle. For random paths, the correct object is

$$\varphi_I(\lambda) = \mathbb{E}e^{i\lambda\pi_I(S(X))},$$

and the nonlinear exponential must be handled before or inside the expectation. The shuffle identity provides exactly this mechanism: the function $e^{i\lambda\pi_I}$ is an infinite linear functional on signatures. Under the exponential-integrability condition Assumption 1, expectation may be exchanged with this linear functional.

The resulting inversion is explicit. The m-th moment of the coordinate signature is obtained from a Taylor coefficient of the path characteristic function evaluated on a finite-dimensional unitary tensor lift $M_I^{[m]}$. Summing these moments with coefficients $(i\lambda)^m/m!$ recovers $\varphi_I(\lambda)$ exactly. Thus, subject to the stated regularity condition, the characteristic function of every coordinate signature random variable is recoverable from the family of unitary path developments, without first recovering the coordinate signature pathwise.

11 Main homogeneous inversion theorem

[Ni: As I have difficulty in compiling the latex of the second version of gpt document. I copied the relevant material to the following sections.]

12 A universal homogeneous word tree

We now construct a skew-Hermitian matrix model that extracts any homogeneous level-N linear functional as a leading Taylor coefficient.

Fix $N \geq 1$. Let

$$H_N = \ell^2(\mathcal{W}_{\leq N})$$

with orthonormal basis $\{e_w : w \in \mathcal{W}_{\leq N}\}$. For words $u, v \in \mathcal{W}_{\leq N}$, let $E_{u,v}$ be the matrix unit sending e_v to e_u. If $a \in \{1, \ldots, d\}$ and $|u| \leq N-1$, write ua for the word obtained by appending a to u. Define

$$B_a^{(N)} = \sum_{|u| \leq N-1} (E_{u,ua} - E_{ua,u}), \qquad a = 1, \ldots, d.$$

[Ni: is this construction related to the shuffle?] Each $B_a^{(N)}$ is skew-Hermitian. Let

$$M_N : \mathbb{R}^d \to \mathfrak{u}(H_N), \qquad M_N(e_a) = B_a^{(N)}.$$

This coefficient map is universal for all level-N homogeneous functionals. If

$$\ell = \sum_{w \in \mathcal{W}_N} c_w \ell_w,$$

set the terminal vector

$$\eta_\ell = \sum_{w \in \mathcal{W}_N} c_w e_w \in H_N.$$

Lemma 4 (Homogeneous word-tree picker)**.** *For every continuous bounded-variation path x,*

$$[\mathrm{Dev}_{tM_N}(x)]_{e_\varnothing,\eta_\ell} = t^N Z_\ell(x) + O_x(t^{N+1}) \qquad (t \to 0).$$

Equivalently,

$$Z_\ell(x) = \frac{1}{N!}\frac{\mathrm{d}^N}{\mathrm{d}t^N}\bigg|_{t=0} [\mathrm{Dev}_{tM_N}(x)]_{e_\varnothing,\eta_\ell}.$$

[Ni: I found Lemma 4 strange. what does η_ℓ mean in $[\mathrm{Dev}_{tM_N}(x)]_{e_\varnothing,\eta_\ell}$? this is correct. please see the below proof by gpt assistant] [Ni: This lemma is key. If this is true, the moment recovery follows just like the cordinate signature case.]

Proof. The signature expansion gives

$$\mathrm{Dev}_{tM_N}(x) = I + \sum_{r=1}^{\infty} t^r \sum_{a_1,\dots,a_r=1}^{d} \pi_{(a_1,\dots,a_r)}(S(x)) B_{a_1}^{(N)} \cdots B_{a_r}^{(N)}.$$

We inspect the coefficient $[B_{a_1}^{(N)} \cdots B_{a_r}^{(N)}]_{e_\varnothing,e_w}$ for $|w| = N$. The product acts on e_w from right to left. Each factor moves by one edge in the rooted word tree, either from a word to its parent or from a word to one of its children. The graph distance from w to the root $\varnothing$ is N. Hence the coefficient is zero for $r < N$.

If $r = N$, every move must be along the unique geodesic from w to $\varnothing$, and every move must decrease word length by one. Write $w = w_1 \cdots w_N$. The rightmost factor must remove the final letter w_N, so it must be $B_{w_N}^{(N)}$; the next factor to the left must be $B_{w_{N-1}}^{(N)}$, and so on. Thus the only nonzero product of length N is

$$B_{w_1}^{(N)} B_{w_2}^{(N)} \cdots B_{w_N}^{(N)}.$$

Along this downward geodesic the relevant summands are $E_{u,ua}$, all with coefficient $+1$. Therefore

$$[B_{a_1}^{(N)} \cdots B_{a_N}^{(N)}]_{e_\varnothing,e_w} = \begin{cases} 1, & (a_1,\dots,a_N) = w, \\ 0, & \text{otherwise.} \end{cases}$$

Taking the coefficient from $e_\varnothing$ to $\eta_\ell = \sum_{|w|=N} c_w e_w$, the coefficient of t^N is

$$\sum_{|w|=N} c_w \pi_w(S(x)) = Z_\ell(x).$$

This proves the claim. □

[Ni: gpt explanation - i think it is correct. Suppose

$$Z_\ell(x) = \sum_{|w|=N} c_w \pi_w(S(x)).$$

Define

$$\eta_\ell = \sum_{|w|=N} c_w e_w.$$

Since $\mathrm{Dev}_{tM_N}(x)$ is linear, we have

$$\mathrm{Dev}_{tM_N}(x)\eta_\ell = \mathrm{Dev}_{tM_N}(x)\left(\sum_{|w|=N} c_w e_w\right) = \sum_{|w|=N} c_w \,\mathrm{Dev}_{tM_N}(x)e_w.$$

Taking the $e_\varnothing$-coefficient gives

$$[\mathrm{Dev}_{tM_N}(x)]_{e_\varnothing,\eta_\ell} = \sum_{|w|=N} c_w [\mathrm{Dev}_{tM_N}(x)]_{e_\varnothing,e_w}.$$

For each word w with $|w| = N$,

$$[\mathrm{Dev}_{tM_N}(x)]_{e_\varnothing,e_w} = t^N \pi_w(S(x)) + O_x(t^{N+1}).$$

Therefore

$$\begin{aligned}[\mathrm{Dev}_{tM_N}(x)]_{e_\varnothing,\eta_\ell} &= \sum_{|w|=N} c_w \left(t^N \pi_w(S(x)) + O_x(t^{N+1})\right) \\ &= t^N \sum_{|w|=N} c_w \pi_w(S(x)) + O_x(t^{N+1}) \\ &= t^N Z_\ell(x) + O_x(t^{N+1}).\end{aligned}$$

]

Remark 7 (A smaller coordinate picker). If $\ell = \ell_I$ is a single coordinate word $I = (i_1, \dots, i_N)$, the full word tree is unnecessary. One may work on $\mathbb{C}^{N+1}$ with basis $e_0, \dots, e_N$ and edges

$$C_a = \sum_{\substack{1\le r\le N \\ i_r = a}} (E_{r-1,r} - E_{r,r-1}).$$

Then

$$[\mathrm{Dev}_{tM_I}(x)]_{e_0,e_N} = t^N \pi_I(S(x)) + O_x(t^{N+1}).$$

This is the path-shaped version of the tree argument. It is more efficient for one coordinate, while the tree model is universal for arbitrary homogeneous linear combinations at level N.

13 Symmetric powers and moment extraction

Let $D_N = \dim H_N = 1 + d + \cdots + d^N$. For $m \ge 0$, let

$$H_{N,m} = \mathrm{Sym}^m H_N.$$

For $m = 0$, this is $\mathbb{C}$. For $m \ge 1$, let ρ_m be the m-th symmetric-power representation of $U(H_N)$, and let $\mathrm{d}\rho_m$ be its Lie algebra representation:

$$\mathrm{d}\rho_m(A) = \sum_{q=1}^{m} I^{\otimes(q-1)} \otimes A \otimes I^{\otimes(m-q)} \quad \text{restricted to } \mathrm{Sym}^m H_N.$$

If A is skew-Hermitian, then $\mathrm{d}\rho_m(A)$ is skew-Hermitian. Define

$$M_{N,m} = \mathrm{d}\rho_m \circ M_N : \mathbb{R}^d \to \mathfrak{u}(H_{N,m}).$$

For $m \ge 1$, set

$$u_0^{(m)} = e_\varnothing^{\otimes m}, \qquad v_\ell^{(m)} = \eta_\ell^{\otimes m},$$

viewed as vectors in $\mathrm{Sym}^m H_N$. For $m = 0$, both vectors are the scalar 1.

Lemma 5 (Covariance under symmetric powers)**.** *For every deterministic continuous bounded-variation path x and every $m \geq 0$,*

$$\mathrm{Dev}_{tM_{N,m}}(x) = \rho_m(\mathrm{Dev}_{tM_N}(x)).$$

Consequently, for $m \geq 1$,

$$[\mathrm{Dev}_{tM_{N,m}}(x)]_{u_0^{(m)}, v_\ell^{(m)}} = \left([\mathrm{Dev}_{tM_N}(x)]_{e_\varnothing, \eta_\ell}\right)^m.$$

Proof. Let Y_s denote the solution of $\mathrm{d}Y_s = Y_s\, tM_N(\,\mathrm{d}x_s)$, $Y_0 = I$, up to time s. Since x and Y have bounded variation, the ordinary product rule applies to tensor powers. The process $\rho_m(Y_s)$ therefore satisfies

$$\mathrm{d}\rho_m(Y_s) = \rho_m(Y_s)\ \mathrm{d}\rho_m(tM_N(\,\mathrm{d}x_s)), \qquad \rho_m(Y_0) = I,$$

which is exactly the development equation driven by $M_{N,m}$. Uniqueness of linear bounded-variation equations gives the first identity.

For the coefficient identity, use

$$\rho_m(Y)\eta_\ell^{\otimes m} = (Y\eta_\ell)^{\otimes m}.$$

Taking the coefficient against $e_\varnothing^{\otimes m}$ gives

$$(e_\varnothing^* Y \eta_\ell)^m.$$

□

Proposition 2 (Homogeneous moment picker)**.** *For every deterministic continuous bounded-variation path x and every $m \geq 1$,*

$$[\mathrm{Dev}_{tM_{N,m}}(x)]_{u_0^{(m)}, v_\ell^{(m)}} = t^{Nm} Z_\ell(x)^m + O_x(t^{Nm+1}).$$

Equivalently,

$$Z_\ell(x)^m = \frac{1}{(Nm)!}\, \frac{\mathrm{d}^{Nm}}{\mathrm{d}t^{Nm}}\bigg|_{t=0} [\mathrm{Dev}_{tM_{N,m}}(x)]_{u_0^{(m)}, v_\ell^{(m)}}.$$

Proof. By Lemmas 4 and 5,

$$\begin{aligned}[\mathrm{Dev}_{tM_{N,m}}(x)]_{u_0^{(m)}, v_\ell^{(m)}} &= \left(t^N Z_\ell(x) + O_x(t^{N+1})\right)^m \\ &= t^{Nm} Z_\ell(x)^m + O_x(t^{Nm+1}).\end{aligned}$$

The derivative formula is the statement that the leading coefficient is the (Nm)-th Taylor coefficient. □

Remark 8 (Dimension)*.* The unsymmetrised tensor construction would use $H_N^{\otimes m}$, of dimension D_N^m. Since only the m-th power of one scalar coefficient is needed, the symmetric power suffices and has dimension

$$\dim \mathrm{Sym}^m H_N = \binom{D_N + m - 1}{m}.$$

For a coordinate word, replacing the tree by the path-shaped picker gives the smaller dimension $\binom{N+m}{m}$.

14 Main homogeneous inversion theorem

Theorem 2 (Homogeneous level-N characteristic function from unitary PCF)**.** *Let X be a measurable random continuous bounded-variation path in $\mathbb{R}^d$. Fix $N \geq 1$, a real homogeneous functional $\ell \in ((\mathbb{R}^d)^{\otimes N})^*$, and $\lambda \in \mathbb{R}$. Suppose Assumption* **??** *holds. Then*

$$\varphi_\ell(\lambda) = \mathbb{E}e^{i\lambda Z_\ell(X)}$$

is recovered from the unitary path characteristic function Φ_X by

$$\boxed{\varphi_\ell(\lambda) = \sum_{m=0}^{\infty} \frac{(i\lambda)^m}{m!} \mu_m(\ell)} \tag{16}$$

with absolute convergence, where $\mu_0(\ell) = 1$ and, for $m \geq 1$,

$$\boxed{\mu_m(\ell) = \frac{1}{(Nm)!} \frac{d^{Nm}}{dt^{Nm}}\bigg|_{t=0} [\Phi_X(tM_{N,m})]_{u_0^{(m)}, v_\ell^{(m)}}.} \tag{17}$$

Here $M_{N,m} : \mathbb{R}^d \to \mathfrak{u}(\mathrm{Sym}^m H_N)$ is the explicit symmetric-power word-tree map constructed above. Thus the m-th coefficient uses a unitary group of order $\binom{D_N+m-1}{m}$, where $D_N = 1 + d + \cdots + d^N$.

The recovery is admissible in the sense of Definition **??***. It uses only expected finite-dimensional unitary developments and exact Taylor coefficients; it does not reconstruct $Z_\ell(X)$ pathwise.*

Proof. Fix $m \geq 1$. Proposition 2 gives, for each deterministic path x,

$$Z_\ell(x)^m = \frac{1}{(Nm)!} \frac{d^{Nm}}{dt^{Nm}}\bigg|_{t=0} [\mathrm{Dev}_{tM_{N,m}}(x)]_{u_0^{(m)}, v_\ell^{(m)}}.$$

For this fixed finite-dimensional matrix map $M_{N,m}$, Assumption **??** gives the local integrability required in Lemma 1 with $q = N$. Hence Taylor coefficients may be passed through expectation:

$$\begin{aligned}\mathbb{E}[Z_\ell(X)^m] &= \frac{1}{(Nm)!} \frac{d^{Nm}}{dt^{Nm}}\bigg|_{t=0} \mathbb{E}\left[[\mathrm{Dev}_{tM_{N,m}}(X)]_{u_0^{(m)}, v_\ell^{(m)}}\right] \\ &= \frac{1}{(Nm)!} \frac{d^{Nm}}{dt^{Nm}}\bigg|_{t=0} [\Phi_X(tM_{N,m})]_{u_0^{(m)}, v_\ell^{(m)}}.\end{aligned}$$

Thus $\mu_m(\ell) = \mathbb{E}[Z_\ell(X)^m]$. Lemma **??** then gives

$$\mathbb{E}e^{i\lambda Z_\ell(X)} = \sum_{m=0}^{\infty} \frac{(i\lambda)^m}{m!} \mathbb{E}[Z_\ell(X)^m]$$

with absolute convergence. Substitution of the moment formula proves (16). □

Corollary 1 (Characteristic function of the level-N signature vector)**.** *Under the same hypotheses, the ordinary finite-dimensional characteristic function of the random tensor $S_N(X)$,*

$$\widehat{\mu}_N(\theta) = \mathbb{E}\exp\big(i\langle\theta, S_N(X)\rangle\big), \qquad \theta \in ((\mathbb{R}^d)^{\otimes N})^*,$$

is recovered from Φ_X at every θ satisfying

$$\mathbb{E}\exp\{\alpha V(X)^N\} < \infty \quad \textit{for some} \quad \alpha > \|\theta\|_1 / N!.$$

Proof. Apply Theorem 2 with $\lambda = 1$ and $\ell = \theta$. □

Corollary 2 (Coordinate case)**.** *Let I be a word of length N. Under the condition*

$$\mathbb{E}\exp\{\alpha V(X)^N\} < \infty \quad \textit{for some} \quad \alpha > |\lambda|/N!,$$

the scalar characteristic function

$$\mathbb{E}e^{\mathrm{i}\lambda\pi_I(S(X))}$$

is recovered from Φ_X. Using the path-shaped picker, the m-th moment query may be made in dimension $\binom{N+m}{m}$.

Remark 9 (The level-zero and level-one cases). For $N = 0$, the signature level is the constant 1. For $N = 1$, a homogeneous functional is an increment functional. If $\ell(x) = \sum_a c_a(x_T^a - x_0^a)$, then there is a one-shot 1-dimensional unitary formula

$$\mathbb{E}e^{\mathrm{i}\lambda Z_\ell(X)} = \Phi_X(M_{\lambda,\ell}), \qquad M_{\lambda,\ell}(e_a) = \mathrm{i}\lambda c_a \in \mathfrak{u}_1.$$

The moment expansion remains valid, but is unnecessary in this linear case.

Remark 10 (Exactness). The theorem is an exact inversion statement in a mathematical oracle model. It assumes exact access to Φ_X and to Taylor coefficients of scalar entries of $\Phi_X(tM)$ at $t = 0$. It is not a finite-sample estimator, and it does not claim numerical stability of high-order differentiation.

15 Shuffle interpretation

Let ℓ be a homogeneous linear functional on level N. Chen's shuffle identity gives, for signatures,

$$\ell(S(x))^m = (\ell^{\sqcup\!\sqcup m})(S(x)),$$

where $\ell^{\sqcup\!\sqcup m}$ is the m-fold shuffle product. Formally,

$$\mathcal{E}_{\ell,\lambda} = \sum_{m=0}^{\infty} \frac{(\mathrm{i}\lambda)^m}{m!}\ell^{\sqcup\!\sqcup m}$$

is the shuffle-exponential of ℓ, and

$$\mathcal{E}_{\ell,\lambda}(S(x)) = e^{\mathrm{i}\lambda Z_\ell(x)}.$$

Theorem 2 can be read as a way to evaluate the expectation of this infinite linear functional from unitary developments.

No completed shuffle algebra is needed for the proof. The required analytic statement is exactly Lemma **??**: under Assumption **??**,

$$\mathbb{E}[\mathcal{E}_{\ell,\lambda}(S(X))] = \sum_{m=0}^{\infty} \frac{(\mathrm{i}\lambda)^m}{m!}\mathbb{E}[(\ell^{\sqcup\!\sqcup m})(S(X))]$$

with absolute L^1-convergence. The symmetric-power word-tree formula then recovers each number $\mathbb{E}[(\ell^{\sqcup\!\sqcup m})(S(X))] = \mathbb{E}[Z_\ell(X)^m]$ from Φ_X.

16 A finite-dimensional obstruction

The homogeneous inversion theorem is finite-dimensional for each moment, but in general it still uses infinitely many moments to recover the scalar characteristic function. The following elementary obstruction explains why a single finite-dimensional unitary coefficient should not be expected for nonlinear homogeneous observables.

Proposition 3 (One-shot obstruction in one dimension)**.** *Let $d = 1$, let $N \geq 2$, and let $\lambda \neq 0$. There do not exist $n \in \mathbb{N}$, $A \in \mathfrak{u}_n$, and vectors $u, v \in \mathbb{C}^n$ such that, for every real x,*

$$u^* \mathrm{e}^{xA} v = \exp\left(\mathrm{i}\lambda \frac{x^N}{N!}\right).$$

Equivalently, even on straight one-dimensional paths, the exponential of a higher homogeneous signature coordinate cannot be a single coefficient of one finite-dimensional unitary development.

Proof. Since A is skew-Hermitian, it is unitarily diagonalizable with eigenvalues $\mathrm{i}\theta_1, \ldots, \mathrm{i}\theta_n$, $\theta_j \in \mathbb{R}$. Therefore $u^* \mathrm{e}^{zA} v$ is an entire exponential polynomial

$$f(z) = \sum_{j=1}^{n} b_j \mathrm{e}^{\mathrm{i}\theta_j z}$$

of order at most one. If the claimed identity held for all real x, the identity theorem would imply

$$f(z) = \exp\left(\mathrm{i}\lambda \frac{z^N}{N!}\right)$$

for every $z \in \mathbb{C}$. The right-hand side is an entire function of order $N \geq 2$, because $\lambda \neq 0$. This contradicts the order bound for finite exponential polynomials. □

Remark 11. The proposition rules out only the strongest possible one-shot representation. It does not preclude finite-dimensional formulas for special laws, nor finite algorithms using several queries. It does show that the infinite analytic step in Theorem 2 is not merely an artefact of the proof.

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